Inference for linear regression models

Activity

Work on the activity (handout), then we will discuss as a class.

Activity

Your parents, worried about the nutritional value of your breakfast, believe that there will be a positive relationship between the amount of sugar in a serving, and the number of calories per serving.

> What is our population model?

$$\text{Calories} = \beta_0 + \beta_1 \text{Sugar} + \varepsilon$$

- contains ε
- no hats
- don't know actual values for the slope & intercept, so we represent them with β_1 & β_0

Activity

```
Coefficients:
            Estimate Std. Error
(Intercept)  87.4277     5.1627
Sugar         2.4808     0.7074
```

How would I interpret the estimated slope?

A one gram increase in sugar per serving is associated with an increase of 2.48 calories per serving, on average

Activity

Your parents, worried about the nutritional value of your breakfast, believe that there will be a positive relationship between the amount of sugar in a serving, and the number of calories per serving.

$$\text{Calories} = \beta_0 + \beta_1 \text{Sugar} + \varepsilon$$

> What are our null and alternative hypotheses?

$H_0: \beta_1 = 0$ (or could do $H_0: \beta_1 \leq 0$)

$H_A: \beta_1 > 0$ (positive relationship)

- in terms of model parameters (e.g. β_1)
- no hats!

Activity

$$H_0 : \beta_1 = 0 \qquad H_A : \beta_1 > 0$$

```
Coefficients:
            Estimate Std. Error
(Intercept)  87.4277     5.1627
Sugar         2.4808     0.7074
```

$df = n - 2$

$= 36 - 2 = 34$

What is our test statistic? What are the degrees of freedom?

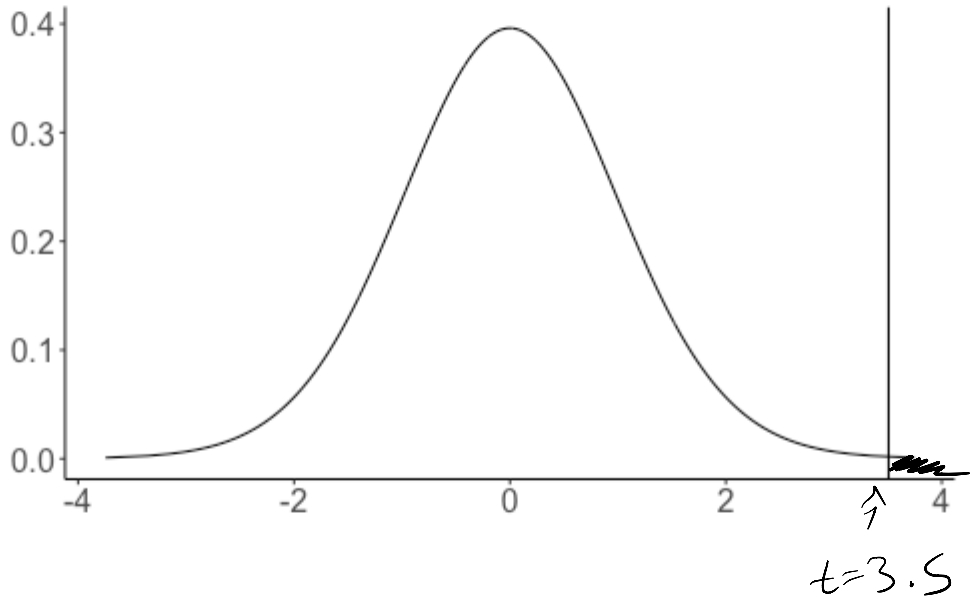
$\hat{\beta}_1$ value under H_0

$$t = \frac{2.4808 - 0}{0.7074} \leftarrow SE_{\hat{\beta}_1}$$

≈ 3.5

$\hat{\beta}_1$ is 3.5 SDs away from hypothesized value of 0 under H_0

Activity



$H_A: \beta_1 > 0$
$\Rightarrow$ right tail

What area under the curve corresponds to my p-value?

A new question

$$\text{Calories} = \beta_0 + \beta_1 \text{Sugar} + \varepsilon$$

```
Coefficients:
            Estimate Std. Error
(Intercept)  87.4277     5.1627
Sugar         2.4808     0.7074
```

Here $\widehat{\beta}_1 = 2.48$. Will $\widehat{\beta}_1$ be exactly equal to the true value β_1?

A new question

$$\text{Calories} = \beta_0 + \beta_1 \text{Sugar} + \varepsilon$$

```
Coefficients:
            Estimate Std. Error
(Intercept)  87.4277     5.1627
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```

> Here $\widehat{\beta}_1 = 2.48$. Will $\widehat{\beta}_1$ be exactly equal to the true value β_1?

No -- our estimate comes from one sample of data, and there is variability from sample to sample.

> How can I account for this variability when reporting an estimate of β_1?

Interval estimates

Point estimate: a single "best guess" value of a parameter. The estimated slope $\widehat{\beta}_1$ is a point estimate of β_1

Interval estimate: a "reasonable range" of possible values for a parameter. A *confidence interval* is an interval estimate

> How would I construct a confidence interval for β_1?

$$\hat{\beta}_1 \pm \underset{\text{critical value}}{t^*} \underbrace{SE_{\hat{\beta}_1}}_{\text{standard error of } \hat{\beta}_1}$$

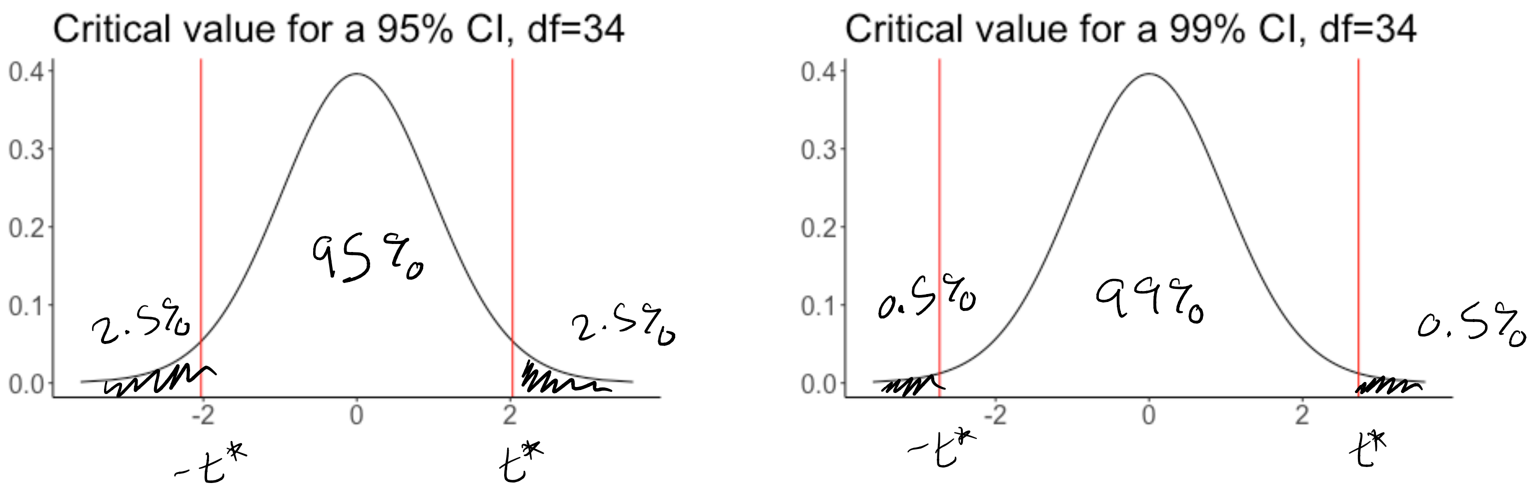
critical value

(how many SDs to go above & below $\hat{\beta}_1$)

standard error of $\hat{\beta}_1$

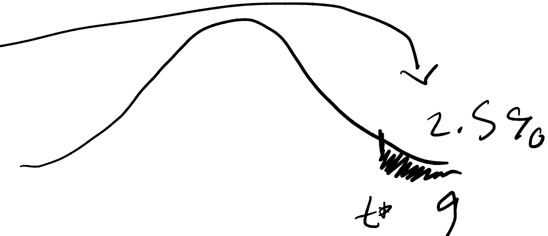
The critical value t^*

To construct a $(1-\alpha)\cdot 100\%$ confidence interval, t^* is the value such that the area in the right tail of the t_{n-2} distribution is $\alpha/2$



As confidence level $\uparrow$, t^* $\uparrow$

Computing t^*



Example: for a 95% confidence interval and 34 degrees of freedom, $t^* = 2.03$

look in right tail

```
qt(0.025, df=34, lower.tail=FALSE)
```

quantiles of t distribution

```
## [1] 2.032245
```

Example: for a 99% confidence interval and 34 degrees of freedom, $t^* = 2.73$

```
qt(0.005, df=34, lower.tail=FALSE)
```

```
## [1] 2.728394
```

Confidence interval

```
Coefficients:
            Estimate Std. Error
(Intercept)  87.4277     5.1627
Sugar         2.4808     0.7074
```

95% confidence interval for β_1:

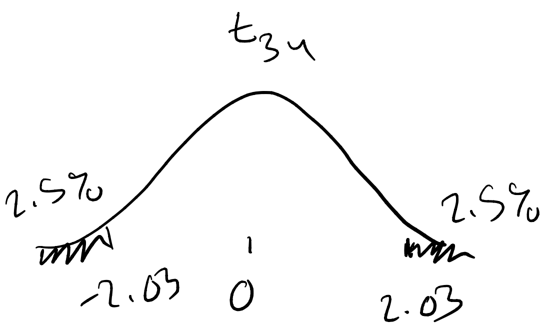
$\downarrow$

$t^* = 2.03$

(34 df)

$\hat{\beta}_1 \pm t^* SE_{\hat{\beta}_1}$

$2.48 \pm 2.03(0.7074) = (1.04, 3.92)$



Confidence interval

```
Coefficients:
            Estimate Std. Error
(Intercept)  87.4277     5.1627
Sugar         2.4808     0.7074
```

95% confidence interval for β_1:

$$2.48 \pm 2.03(0.71) = (1.04, 3.92)$$

> How do I interpret this confidence interval?

Confidence interval

```
Coefficients:
            Estimate Std. Error
(Intercept)  87.4277     5.1627
Sugar         2.4808     0.7074
```

95% confidence interval for β_1:

$$2.48 \pm 2.03(0.71) = (1.04, 3.92)$$

Interpretation: We are 95% confident that a one-gram increase in sugar per serving of breakfast cereal is associated with an increase of between 1.04 and 3.92 calories per serving.

> What does it mean to be "95% confident"?

If we repeat the process and take many samples, and calculate a 95% CI for each sample, then we expect 95% of the intervals to contain the β_1

Activity

Work on the activity (handout), then we will discuss as a class.

Activity

Data on 116 sparrows. Want to make a 98% confidence interval.

```
qt(..., df = ..., lower.tail=FALSE)
```

> What do I fill in to calculate the critical value t^*?

Activity

Data on 116 sparrows. Want to make a 98% confidence interval.

```
qt(0.01, df = 114, lower.tail=FALSE)
```

```
## [1] 2.359504
```

Activity

```
              Estimate Std. Error
(Intercept)  8.75465    1.39601
Weight       1.31341    0.09756
```

$t^* = 2.36$

> What is my 98% confidence interval?

Activity

```
              Estimate Std. Error
(Intercept)  8.75465    1.39601
Weight       1.31341    0.09756
```

$t^* = 2.36$

$$1.31 \pm 2.36(0.098) = (1.08, 1.54)$$

> Your friend claims that there is a 2% chance that the true slope β_1 is greater than 1.54 or less than 1.08. Is your friend correct?